Inputs

**BOLD (COI) or UNITE (ITS)**
reference barcodes, BINs or SHs

**Open Tree**
topology

**TimeTree**
divergence times

from published syntheses
from barcodes
related genera, for rooting only

age from TimeTree
age interpolated (pure birth)

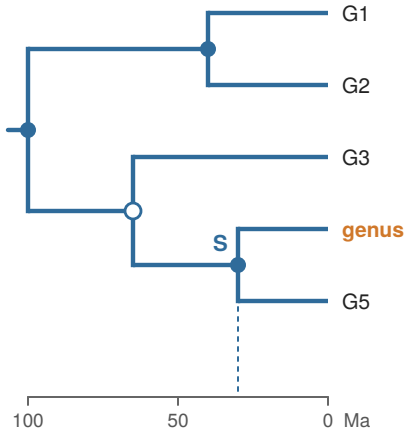
## (i) Reference units

one representative sequence per BIN

filters: length, ambiguous bases, reading frame (COI)
genus = majority identification

## (ii) Dated backbone

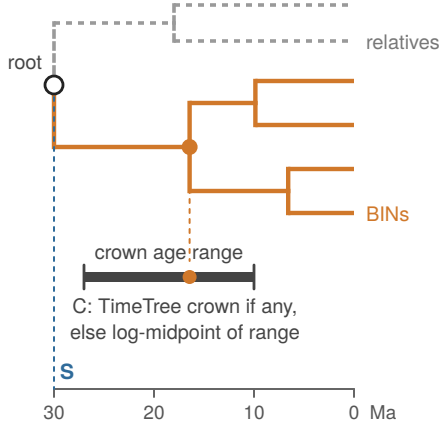
one tip per genus, ages in Ma




S: stem age of the genus

## (iii) Dated subtree per genus

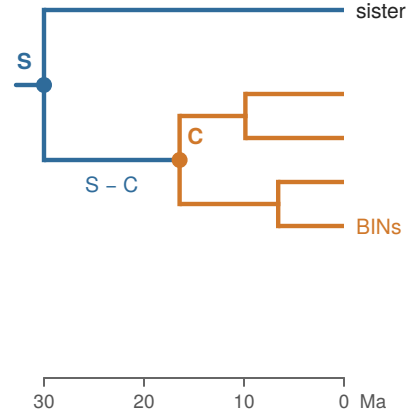
barcode tree, rooted on related genera




min: BIN divergence ÷ rate
max: 0.9 S and TimeTree crown

## (iv) Grafting

genus tip replaced by its subtree




no rescaling: both parts in Ma;
genera run in parallel

## Samples: placement and diversity

metabarcoding sequence (ASV) → best match among reference sequences

- **≥ 97%** assigned to that BIN
- **80–97%** placed on the tree by nearest BIN
- **< 80%** left out and reported

**Diversity in Ma of history**

Faith's PD, mean pairwise distance; between samples: UniFrac, PhyloSor